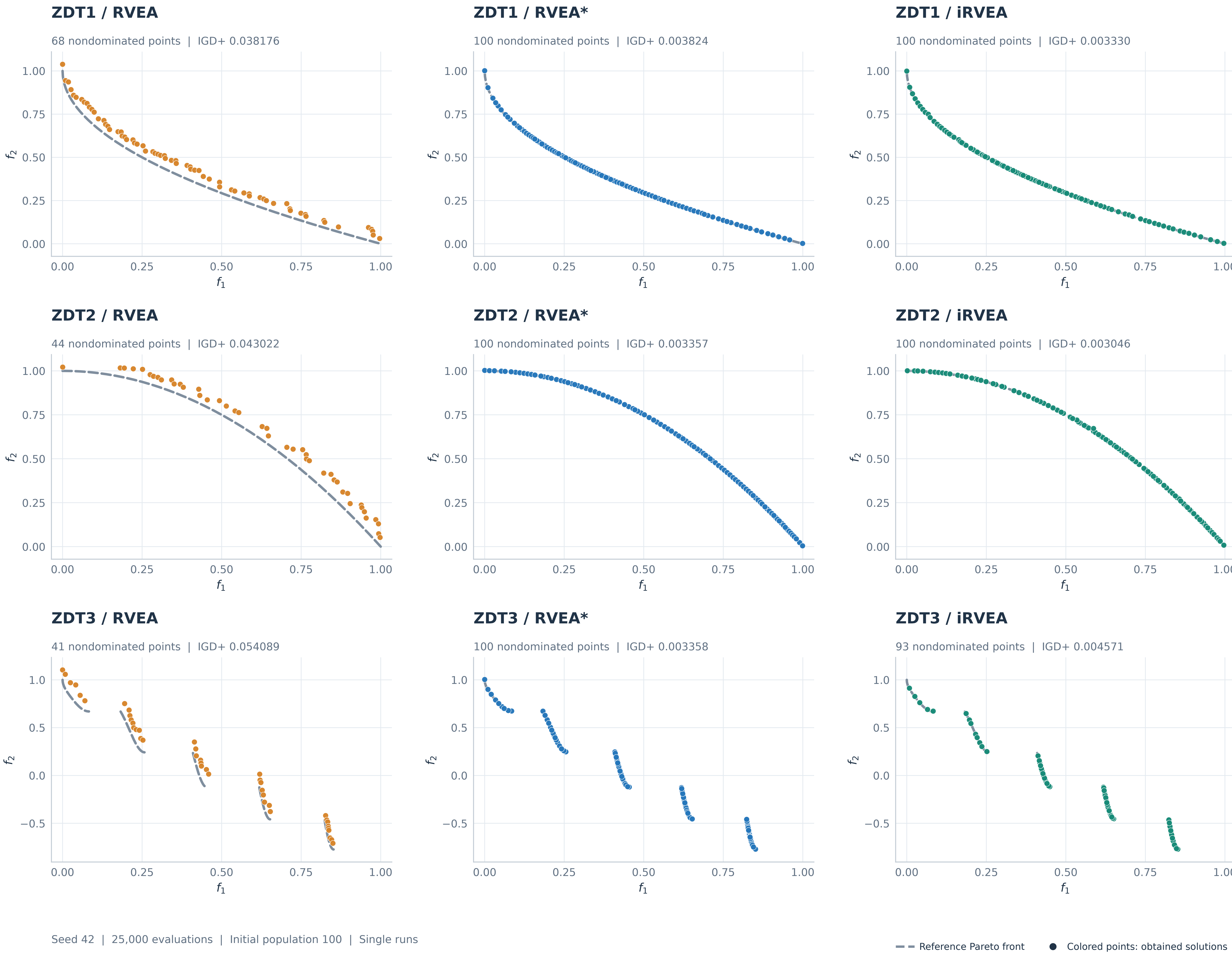
RVEA family / three simpler two-objective problems

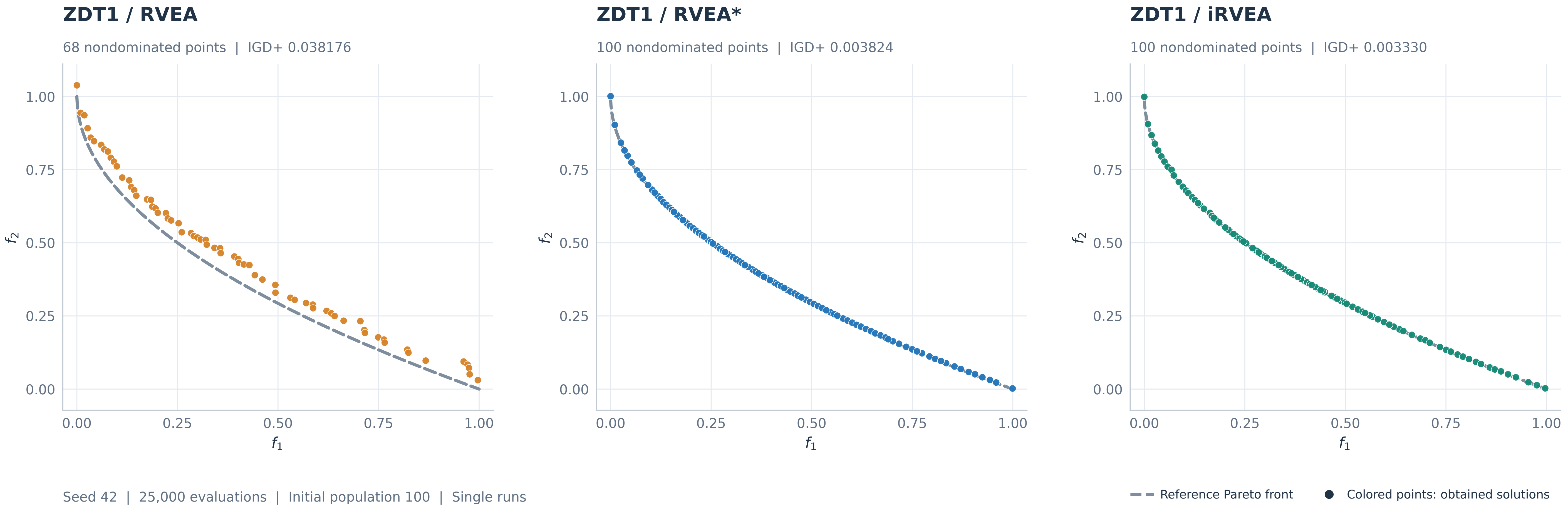
ZDT1: convex | ZDT2: concave | ZDT3: disconnected. 30 decision variables in each problem.



Seed 42 | 25,000 evaluations | Initial population 100 | Single runs

ZDT1 / convex front

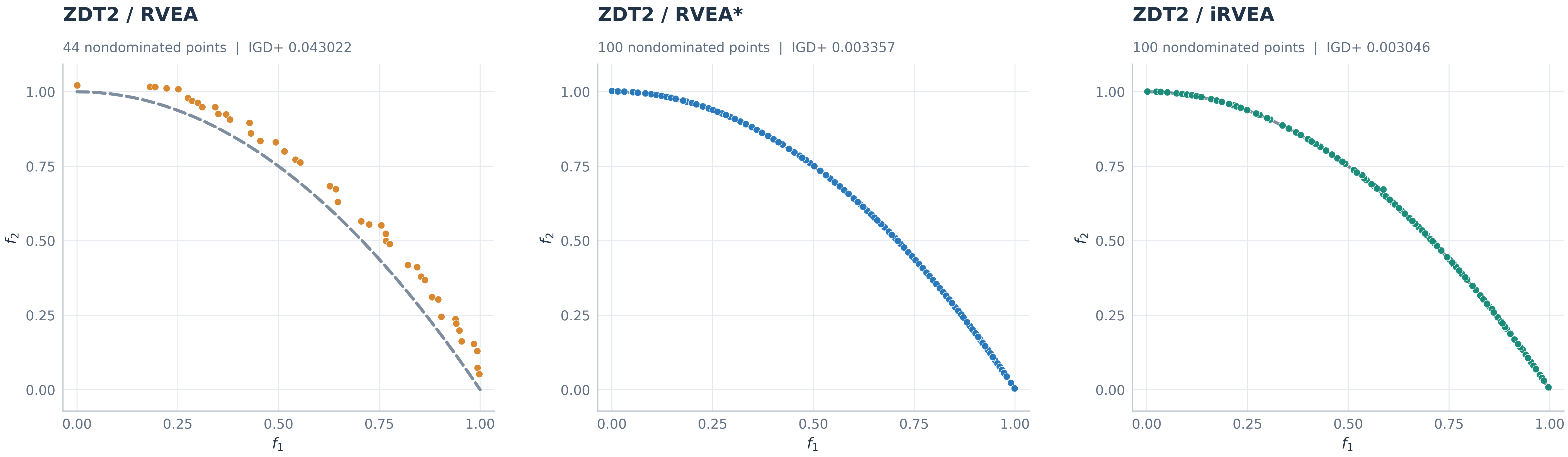
Two objectives, 30 variables. Identical axes across algorithms; lower IGD+ is better.



Seed 42 | 25,000 evaluations | Initial population 100 | Single runs

ZDT2 / concave front

Two objectives, 30 variables. Identical axes across algorithms; lower IGD+ is better.



Seed 42 | 25,000 evaluations | Initial population 100 | Single runs

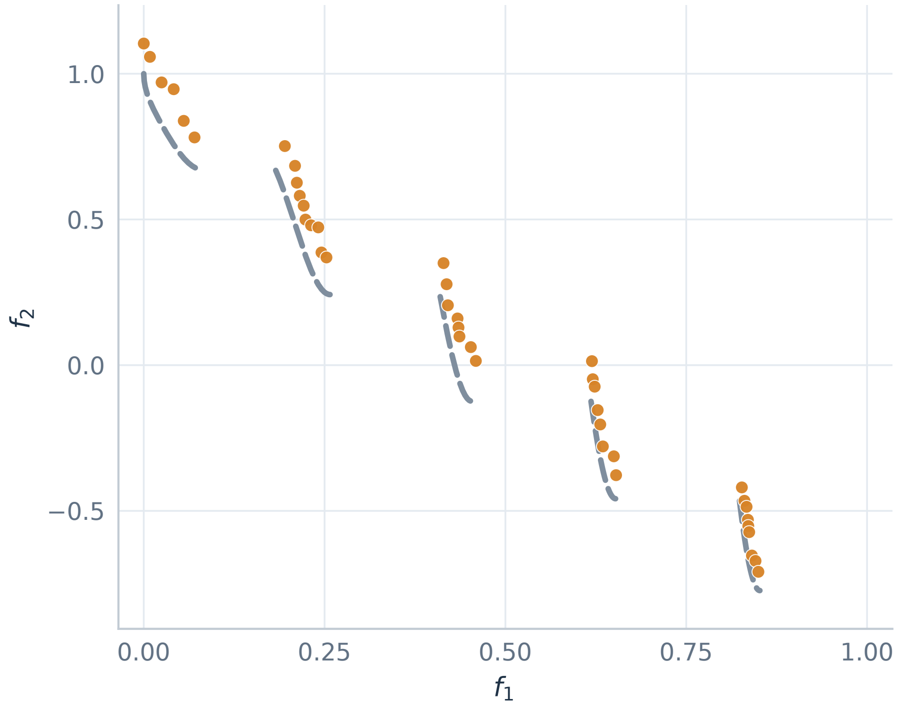
Reference Pareto front   Colored points: obtained solutions

ZDT3 / disconnected front

Two objectives, 30 variables. Identical axes across algorithms; lower IGD+ is better.

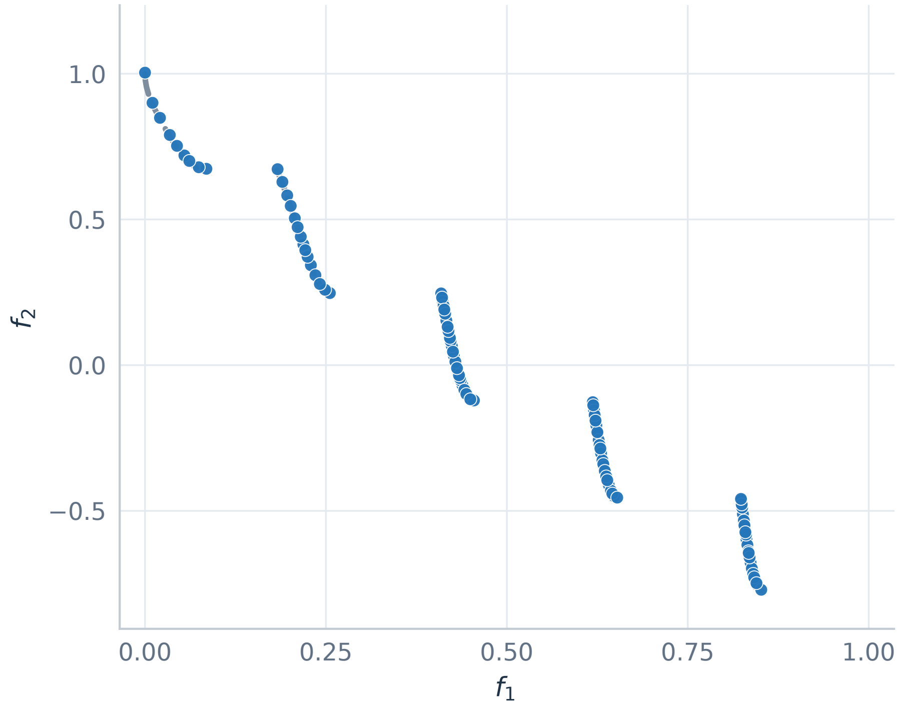
ZDT3 / RVEA

41 nondominated points | IGD+ 0.054089



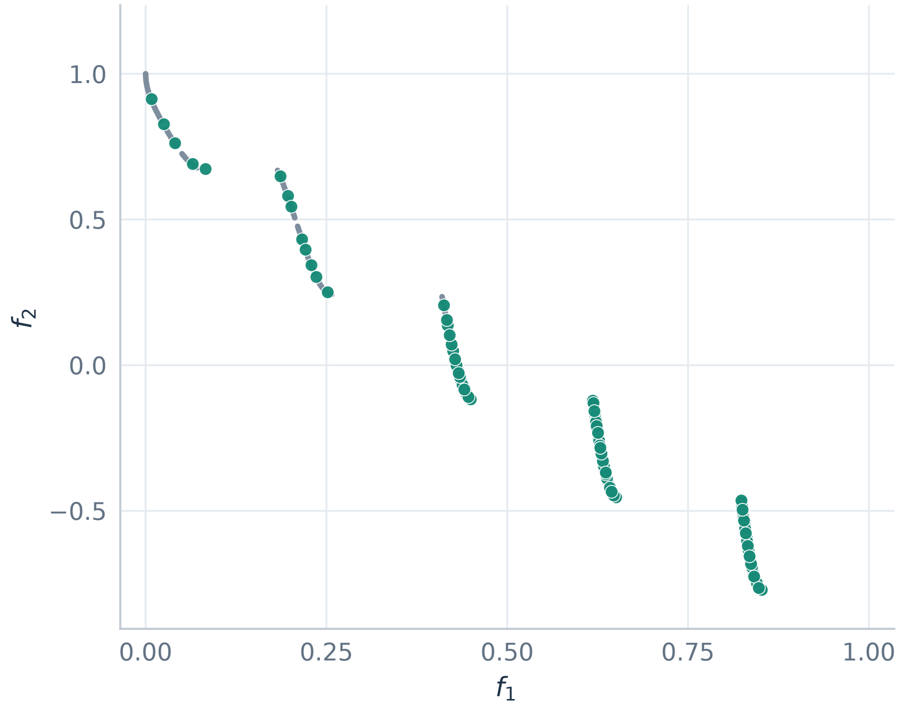
ZDT3 / RVEA*

100 nondominated points | IGD+ 0.003358



ZDT3 / iRVEA

93 nondominated points | IGD+ 0.004571



Seed 42 | 25,000 evaluations | Initial population 100 | Single runs

Reference Pareto front

Colored points: obtained solutions